Last updated: November 2, 2026

Donghyun Lee, Ph.D.

Division of Social Science & AI, Hankuk University of Foreign Studies
107, Imun-ro, Dongdaemun-gu, Seoul, 02450, Republic of Korea
Tel: +82-2-2173-3054 | donghyun.lee@hufs.ac.kr | donghyunlee-lab.github.io

ACADEMIC PROFILE

Trained in technology and innovation management, I study the reliability of AI forecasts for environmental monitoring and infectious-disease surveillance. My publications also include research on green-electricity policy and collaborative studies of technology acceptance and policy preferences. This work draws on experience at the Korea Environment Institute and research projects with public agencies. My developing research agenda examines how public agencies evaluate and use technical evidence in their decisions.

RESEARCH INTERESTS

Current work: Forecast reliability, uncertainty quantification, explanation robustness, and transfer learning, and edge-case validation; technology acceptance and policy preferences.
Developing agenda: Evaluation and use of AI evidence by public agencies; validity of LLM-based synthetic responses for policy research.

ACADEMIC AND POLICY RESEARCH APPOINTMENTS

Hankuk University of Foreign Studies — *Seoul, Republic of Korea*
Associate Professor, Division of Social Science & AI — Sep. 2025 – Present

Tech University of Korea — *Siheung-si, Republic of Korea*
Associate/Assistant Professor, Department of Business Administration — Mar. 2017 – Aug. 2025

Korea Environment Institute — *Sejong, Republic of Korea*
Associate Research Fellow (부연구위원), Big Data Research Team — Jul. 2016 – Feb. 2017

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, Republic of Korea*
Ph.D. in Engineering, Technology and Innovation Management (College of Business) — Feb. 2012 – Aug. 2016

- *Dissertation*: "Essays on big data analytics in industrial research: Green energy policy and patent litigation."
- *Advisor*: Prof. Minki Kim

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, Republic of Korea*
Master of Engineering, Technology and Innovation Management (College of Business) — Sep. 2009 – Aug. 2011

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, Republic of Korea*
Bachelor of Engineering (Information and Communication Engineering; IT Business minor) — Feb. 2005 – Aug. 2009

JOURNAL ARTICLES

* *denotes corresponding author.*

[1] J. Y. Kim, **D. Lee*** (2026). "Uncovering reliability blind spots in transportation AI: An edge-case-based validation approach for transportation systems." *Scientific Reports*. Published online Aug. 1, 2026. doi:10.1038/s41598-026-62679-w.

[2] **D. Lee*** (2026). "Heterogeneous transfer learning for robust infectious disease forecasting: A data-centric approach." *Expert Systems with Applications*, 332 (2027), 133728. Published online July 22, 2026. doi:10.1016/j.eswa.2026.133728.

[3] **D. Lee***, H. Jeon (2025). "Reinforced explainable AI for algal bloom forecasting under climate change: A multi-run class activation mapping (CAM) approach." *Journal of Cleaner Production*, 529, 146805.

[4] M. Oh, G. Lee, **D. Lee**, J. Ahn, J. Shin (2025). "Enhancing Small-Business Ecosystems Through Local Currency Designs in South Korea: A Choice Experiment Approach." *Systems*, 13(5), 339.

[5] S. Park, J. Ryu, **D. Lee**, J. Shin (2025). "How Can We Improve the Consumer Acceptance Level for Disposers Considering Regional Characteristics?" *Water*, 17(4), 493.

[6] **D. Lee***, B. Lee (2024). "Building reliable AI for quantifying uncertainty in particulate matter predictions with deep learning." *Journal of Cleaner Production*, 473, 143457.

[7] M. Kim, **D. Lee*** (2023). "Why Uncertainty in Deep Learning for Traffic Flow Prediction Is Needed." *Sustainability*, 15(23), 16204.

[8] **D. Lee***, M. Kim, B. Lee, S. Chae, S. Kwon, S. Kang (2022). "Integrated explainable deep learning prediction of harmful algal blooms." *Technological Forecasting and Social Change*, 185, 122046.

[9] S. Chae, J. Shin, S. Kwon, S. Lee, S. Kang, **D. Lee*** (2021). "PM10 and PM2.5 real-time prediction models using an interpolated convolutional neural network." *Scientific Reports*, 11(1), 11952.

[10] S. Chae, S. Jang, S. Lee, **D. Lee*** (2020). "Complex System Analysis of Korean Peninsula Earthquake Data." *Scientific Reports*, 10(1), 2686.

[11] S. Huh, H. Lee, J. Shin, **D. Lee**, J. Jang (2018). "Inter-fuel substitution path analysis of the Korea cement industry." *Renewable and Sustainable Energy Reviews*, 82, 4091–4099.

[12] J. Shin, S. Kang, **D. Lee***, B. Hong (2018). "Analysing the failure factors of eco-friendly home appliances based on a user-centered approach." *Business Strategy and the Environment*, 27(8), 1399–1408.

[13] S. Chae, S. Kwon, **D. Lee*** (2018). "Predicting Infectious Disease Using Deep Learning and Big Data." *International Journal of Environmental Research and Public Health*, 15(8), 1596.

[14] S. Lee, **D. Lee*** (2018). "Improved Prediction of Harmful Algal Blooms in Four Major South Korea's Rivers Using Deep Learning Models." *International Journal of Environmental Research and Public Health*, 15(7), 1322.

[15] Y. Yoon, H. Zo, M. Choi, **D. Lee**, H. Lee (2018). "Exploring the dynamic knowledge structure of studies on the Internet of things: Keyword analysis." *ETRI Journal*, 40(6), 745–758.

[16] **D. Lee**, S. Kang, J. Shin (2017). "Using Deep Learning Techniques to Forecast Environmental Consumption Level." *Sustainability*, 9(10), 1894.

[17] **D. Lee**, S. Kang, J. Shin (2017). "Determinants of Pro-Environmental Consumption: Multicountry Comparison Based upon Big Data Search." *Sustainability*, 9(2), 183.

[18] **D. Lee**, J. Kim, J. Shin (2017). "Scale-free network analysis of big data for patent litigation cases in the United States." *Journal of the Korean Physical Society*, 70(4), 431–435.

[19] **D. Lee**, M. Kim, J. Lee (2016). "Adoption of green electricity policies: Investigating the role of environmental attitudes via big data-driven search-queries." *Energy Policy*, 90, 187–201.

[20] **D. Lee**, H. Lee, M. Choi (2016). "Examining the Relationship Between Past Orientation and US Suicide Rates: An Analysis Using Big Data-Driven Google Search Queries." *Journal of Medical Internet Research*, 18(2), e35.

WORKSHOP PAPERS (NON-ARCHIVAL)

[1] H. Jeon, **D. Lee*** (2026). "From next-action fidelity to rollout validity: A human-grounded audit of LLM user simulators in repeated social interaction." *NeurIPS 2026 Workshop on Grounded User Simulation for Model Evaluation and Training (UserSim)*, Paris, Dec. 2026. Peer-reviewed.

TEACHING INTERESTS

Areas of interest: Policy analysis; technology and society; evidence, uncertainty, and public decisions.
My teaching interests center on helping students interpret evidence, compare policy alternatives, and explain the limits of their recommendations.

TEACHING EXPERIENCE

Rating = end-of-term student course evaluation score (out of 5). Tech University of Korea figures are means over the listed sections (university records, Spring 2017–Spring 2024), each shown with the university undergraduate average for the same sections. KOCW figures are ratings by platform users.

Hankuk University of Foreign Studies *Seoul, Republic of Korea*
Associate Professor Sep. 2025 – Present

- **Understanding Technology Development Research** (M04112101). Technology management and high-tech industry; technology development methodologies (consumer preference, market analysis); commercialization of new technologies. Fall 2025 (Rating: 4.23/5, Univ. Undergrad Avg: 4.61).
- **Industrial Data Visualization** (M04111101). Data preprocessing with Python (NumPy, Pandas); visualization with Matplotlib, Streamlit, GitHub for interactive dashboards. Fall 2025 (Rating: 4.7/5, Univ. Undergrad Avg: 4.61).
- **Fundamentals of Social Data Programming** (M04103201). Fundamental concepts of programming using Python: data types, operators, conditionals, loops, functions, classes, modules, packages, exception handling. Fall 2025 (Rating: 4.8/5, Univ. Undergrad Avg: 4.61).
- **Object-Oriented Programming** (M04107101). Principles and practical applications of Object-Oriented Programming (OOP) through Python; object-oriented design skills for social science data modeling. Spring 2026 (Rating: 4.7/5, Univ. Undergrad Avg: 4.57).
- **Deep Learning and Applications** (M04114101). Theory and practice of artificial neural networks from individual neurons to modern deep learning architectures (CNN, LSTM, etc.); implementing core mechanisms from scratch with NumPy, then building practical models with PyTorch for data analysis. Spring 2026 (Rating: 4.45/5, Univ. Undergrad Avg: 4.57).

Tech University of Korea *Siheung-si, Republic of Korea*
Associate/Assistant Professor Mar. 2017 – Aug. 2025

11 distinct courses across undergraduate and graduate levels (60+ sections). Across all 59 evaluated undergraduate sections: mean rating 4.44/5 (Univ. Undergrad Avg: 4.33). Representative courses:

- **Management Capstone Design I/II/Planning** (Undergraduate research). 20 sections, Fall 2017–Spring 2024 (mean rating: 4.48/5, Univ. Undergrad Avg: 4.34).
- **Graduate: New Technology Management** (Technology management theories). 1 semester.
- **Graduate: Information Strategy and Management Innovation** (MIS, big data, AI). 2 semesters.
- **Big Data Analysis** (R programming, statistics, visualization). 13 sections, Spring 2018–Spring 2024 (mean rating: 4.34/5, Univ. Undergrad Avg: 4.33).

- **Programming and Data Analysis** (Python, OOP, data preprocessing). 6 sections, Fall 2021–Fall 2023 (mean rating: 4.58/5, Univ. Undergrad Avg: 4.44).
- **Fundamentals of Programming** (Java). 8 sections, Spring 2017–Spring 2024 (mean rating: 4.60/5, Univ. Undergrad Avg: 4.30).
- **Advanced IT Service Techniques** (Deep learning/ML with Python). 5 sections, Fall 2017–Fall 2021 (mean rating: 4.44/5, Univ. Undergrad Avg: 4.27).

KOCW (Korea OpenCourseWare) *South Korea, kocw.net*
Official Public Online Course

- **Python Programming**. Fundamental programming with Python (conditionals, loops). Online since Spring 2020. 26 lectures. KOCW rating: 4.4/5. Link.
- **Python Programming 2**. Object-oriented Python, NumPy, pandas. Online since Fall 2020. 26 lectures. Link.
- **Big Data Analysis**. R programming and data analysis for big data. Online since Fall 2019. 26 lectures. KOCW rating: 4.5/5. Link.
- **Fundamentals of Programming**. Java, core programming concepts. Online since Spring 2020. 26 lectures. KOCW rating: 5.0/5. Link.

K-DIGITAL Training Online Course *Siheung-si, Republic of Korea*
Instructor

- **Deep Learning**. Principles of deep learning; implementing models with NumPy and Python (without TensorFlow/PyTorch). Jun. 2021 – Aug. 2023 (Cohorts 1–6). 20 lectures.

POLICY SERVICE AND ACADEMIC LEADERSHIP

Government Advisory

National Sustainable Development Research Center (Korea Environment Institute) *South Korea*
Member, Pre-Review Committee for Draft Government Plans under the Korean SDGs (K-SDGs) Sep. 2026 – Present

Ministry of Environment *South Korea*
Committee Member, Central Environmental Policy Committee, Environmental Economics Division Jul. 2022 – Jul. 2024

Editorial Boards

International Telecommunications Policy Review (KCI) *South Korea*
Editorial Board Member 2026 – Present

Journal of Innovation & Knowledge *Amsterdam, Netherlands*
Editorial Review Board Member Apr. 2025 – Present

Professional Societies

Korea Environmental Policy and Administration Society (KEPAS) *South Korea*
Chair, AI and Digital Committee Aug. 2026 – Present

Korea Society for Innovation Management & Economics (KOSIME) *South Korea*
Vice Chair, Special Committee on Energy Industry & Technology Policy 2026 – Present

Academic Leadership

Tech University of Korea

- Director, SW Education Center (Sep. 2022 – Jul. 2024)
- Chief Professor, Venture Entrepreneurship Integration Major (Nov. 2019 – Sep. 2022)
- Chief Professor, Smart Factory Integration Major (Sep. 2018 – Aug. 2019)

PUBLIC-SECTOR RESEARCH REPORTS

[1] J. Kang, **D. Lee**, H. Jeon (2026). "Predicting the Spread of Avian Influenza in Wild Birds (III)." Research Report, National Institute of Wildlife Disease Control and Prevention (NIWDCP).

[2] **D. Lee**, J. Mo, J. Shin, H. Jeon, H. M. Nguyen (2025). "Predicting the Spread of Avian Influenza in Wild Birds (II)." Research Report, NIWDCP.

[3] **D. Lee**, J. Shin, M. Kim (2024). "Predicting the Spread of Avian Influenza in Wild Birds." Research Report, NIWDCP.

[4] S. Kang, **D. Lee**, G. Ko, D. Jin, H. Hong, D. Kim, S. Kang (2023). "Big Data Analysis: Application to Environmental Research and Service VII." Policy Research Report, Korea Environment Institute.

[5] S. Kang, **D. Lee**, J. Pyo, G. Ko, Y. Cho, G. Han, D. Jin, S. Kang (2022). "Big Data Analysis: Application to Environmental Research and Service VI." Policy Research Report, pp. 1-108, Korea Environment Institute.

[6] H. Hong, **D. Lee**, G. Ko, D. Jin, S. Kang, S. Kang, D. Kim (2021). "Big Data Analysis: Application to Environmental Research and Service V." Policy Research Report, pp. 1-195, Korea Environment Institute.

[7] S. Kang, **D. Lee**, G. Ko, D. Jin, H. Hong, G. Han, S. Kang, D. Kim (2019). "Big Data Analysis: Application to Environmental Research and Service III." Policy Research Report, pp. 1-288, Korea Environment Institute.

[8] S. Kang, **D. Lee**, K. Jang, D. Jin, H. Hong, G. Han, J. Kim, S. Kang, D. Kim (2018). "Big Data Analysis: Application to Environmental Research and Service II." Policy Research Report, pp. 1-271, Korea Environment Institute.

[9] S. Kang, **D. Lee**, K. Jang, D. Jin, H. Hong, G. Han, J. Kim, S. Kang, D. Kim, E. Jung (2017). "Big Data Analysis: Application to Environmental Research and Service." Policy Research Report, pp. 1-465, Korea Environment Institute.

[10] D. Kim, S. Song, S. Park, J. Kim, **D. Lee**, S. Jo (2017). "A Study on Expansion of Pro-Environmental Consumption Using Social Economy." Policy Research Report, pp. 1-246, Korea Environment Institute.

[11] S. Kwak, J. Shin, **D. Lee**, S. Kang (2016). "A Study on Promoting Environmental Consumption by Developing the Environmental Attitude-Behavior Model." Policy Research Report, pp. 1-226, Korea Environment Institute.

BOOK CHAPTERS

[1] S. Kang, **D. Lee*** (2022). "Use of artificial intelligence for predicting infectious disease." In P. Keikhosrokiani (Ed.), *Big Data Analytics for Healthcare: Datasets, Techniques, Life Cycles, Management, and Applications*, Elsevier, pp. 153–163. ISBN: 978-0-323-91907-4.

FUNDED RESEARCH PROJECTS

Current Projects

[1] **Institute of Information & Communications Technology Planning & Evaluation (IITP)** *Apr. 2026 – Dec. 2033*
Principal Investigator, "Development of an Ultra-Fast, High-Reliability Physics-AI Hybrid Disaster Response Technology Based on MambaFlow Matching" (Digital Columbus Project).

[2] **National Research Foundation of Korea (NRF)** *Mar. 2026 – Feb. 2029*
Principal Investigator, "Development of Physics-AI Integrated Dynamic Graph Meta-Learning Based Algal Bloom Prediction Technology for Overcoming Data Scarcity" (Outstanding Young Scientist Grant).

Past Projects

[3] **National Institute of Wildlife Disease Control and Prevention (NIWDCP)** *Aug. 2025 – May 2026*
Co-Investigator, "Wild Bird Avian Influenza Spread Prediction Research (III)"

[4] **National Institute of Wildlife Disease Control and Prevention (NIWDCP)** *Jul. 2024 – May 2025*
Principal Investigator, "Wild Bird Avian Influenza Spread Prediction Research (II)"

[5] **National Research Foundation of Korea** *Mar. 2020 – Feb. 2025*
Principal Investigator, "Simulation of Infectious Disease Prediction Using Transfer Learning Deep Learning"

[6] **Tech University of Korea** *Oct. 2023 – Oct. 2024*
Principal Investigator, "Development of Advanced Algal Bloom Prediction Model Using Explainable AI" (1st Place, Research Competition)

[7] **Office of Strategic R&D Planning** *Jul. 2024 – Sep. 2024*
Principal Investigator, "Utilizing Big Data and Large Language Models for Analyzing Trends and Demand in the Industrial Workforce Market"

[8] **National Institute of Wildlife Disease Control and Prevention (NIWDCP)** *Jul. 2023 – May 2024*
Principal Investigator, "Wild Bird Avian Influenza Spread Prediction Research"

[9] **National Research Foundation of Korea** *Nov. 2023 – Mar. 2024*
Co-Investigator, "Development of Performance Indicators and Management System for LAMP Project"

[10] **Kookmin University** *Oct. 2023 – Dec. 2023*
Principal Investigator, "Development of ML-Based Performance Prediction Model for Seawater Desalination Plants"

[11] **Kookmin University** *Sep. 2023 – Nov. 2023*
Principal Investigator, "Development of ML-Based Prediction Model for Water Quality and Energy Consumption in Sewage Treatment Plants"

[12] **Korea Testing Laboratory (KTL)** *Jun. 2023 – Oct. 2023*
Principal Investigator, "Test Data Analysis for Remaining Life Measurement of Electric Vehicle Lithium-Ion Batteries"

[13] **Korea Environment Institute** *Mar. 2023 – Oct. 2023*
Principal Investigator, "Construction of Automated Data Pipeline for Algal Bloom Prediction Using Deep Learning for MLOps"

[14] **National Institute of Wildlife Disease Control and Prevention (NIWDCP)** *Aug. 2022 – Nov. 2022*
Principal Investigator, "Establishment of a Rapid Response System for Avian Influenza"

[15] **Korea Testing Laboratory (KTL)** *Jun. 2022 – Oct. 2022*
Principal Investigator, "Research on Application of Functional Safety Standards in HVAC Using AI"

[16] **Korea Environment Institute** *Apr. 2022 – Oct. 2022*
Principal Investigator, "Development of a New Fine Dust Prediction Module Based on Explainable AI"

[17] **Exabyte Inc.** *Feb. 2022 – Jul. 2022*
Principal Investigator, "Development of Object Detection Model Using Deep Learning"

[18] **Korea Environment Institute** *Mar. 2021 – Nov. 2021*
Principal Investigator, "Deep Learning-Based Prediction of Fine Dust Concentration and Uncertainty"

[19] **Korea Institute of Design Promotion** *Jun. 2020 – Dec. 2020*
Co-Investigator, "Support for Multidisciplinary Education Courses Led by Design"

[20] **National Research Foundation of Korea** *Mar. 2017 – Feb. 2020*
Principal Investigator, "Application of Deep Learning Technology in Environmental and Industrial Research"

[21] **Korea Environment Institute** *Feb. 2019 – Nov. 2019*
Principal Investigator, "Deep Learning-Based Integrated Prediction of Environmental Pollution"

[22] **Jeongseok Logistics Academic Foundation** *Jan. 2019 – Dec. 2019*
Principal Investigator, "Deep Learning-Based Traffic Prediction for Logistics Optimization"

[23] **Tech University of Korea** *Jul. 2018 – Nov. 2018*
Principal Investigator, "Quantitative Analysis of Factors Influencing Dropout Rates Based on Big Data"

[24] **National Research Foundation of Korea** *Mar. 2018 – Feb. 2019*
Principal Investigator, "Application of Deep Learning Technologies — Initial Innovation Laboratory"

[25] **Korea Environment Institute** *Mar. 2018 – Nov. 2018*
Principal Investigator, "Fine Dust Prediction Using Convolutional Neural Networks"

[26] **Korea Institute for Advancement of Technology (KIAT)** *Mar. 2018 – Feb. 2022*
Co-Investigator, "Smart Factory Operation Design and Professional Personnel Training Project"

[27] **National Research Foundation of Korea** *Mar. 2018 – Aug. 2019*
Co-Investigator, "LINC+ Leading University for Innovation in the 4th Industrial Revolution"

[28] **Korea Environment Institute** *Jul. 2017*
Principal Investigator, "Natural Language Analysis and Advisory of Water Source Related Complaint Data"

[29] **National Research Foundation of Korea** *Jul. 2017 – Jun. 2019*
Principal Investigator, "Predicting Epidemic Spread Using Deep Learning Analysis of Query Big Data"

[30] **Korea Environment Institute** *Apr. 2017 – Sep. 2017*
Principal Investigator, "Analysis of Eco-Friendly Consumption Using Social Networking Services (SNS)"

[31] **Tech University of Korea** *Apr. 2017 – Mar. 2019*
Principal Investigator, "Application of Deep Learning Technology in Social Science Research"

[32] **Korea Environment Institute** *Mar. 2017 – Nov. 2017*
Principal Investigator, "Environmental Risk Prediction Using Deep Learning"

INDUSTRY EXPERIENCE

AI Korea Inc. *Incheon, Republic of Korea*
CEO and Founder Dec. 2022 – Aug. 2025,
Feb. 2026 – Present

- Developed AI forecasting solutions for environmental monitoring through applied system development.

CONFERENCE PRESENTATIONS

[1] H. Jeon, **D. Lee*** (presenter). "Country-Level Forecasting of Highly Pathogenic Avian Influenza Outbreaks Using International Outbreak Notifications: Supporting Risk-Based Surveillance through Subtype-Linked Transmission Pathways" (국제 발생 신고 자료를 이용한 고병원성 조류인플루엔자 국가별 발생 예측: 아형 연계 전파 경로 중심의 위험기반 예찰 지원). *2026 Fall Conference of the Korean Society for Zoonoses*, Sep. 11, 2026.

[2] "On-Device Agentic AI Framework for Predicting Spatial Jumps of Emerging Infectious Diseases: A Lumpy Skin Disease (LSD) Case Study" (On-Device Agentic AI 프레임워크를 활용한 신종 감염병의 공간적 점프 예측: 럼피스킨병(LSD) 사례 연구). *Korean Society for Zoonoses*, Oct. 2025.

[3] "Explainable AI-Based Avian Influenza Outbreak Prediction Considering Uncertainty" (불확실성을 고려한 설명 가능한 인공지능 기반 조류 인플루엔자 발생 예측 연구). *Joint Conference on Economics*, Feb. 2025.

[4] "Analysis of Industrial Workforce Market Trends and Demand Using Big Data and Large Language AI Models" (채용 빅데이터와 대형 언어 인공지능 모델을 활용한 산업 인력 시장 동향 및 수요 분석). *Summer Conference of the Korea Technology Innovation Society*, Aug. 2024.

[5] "Spatiotemporal Modeling and Factor Analysis for Predicting Avian Influenza Spread" (조류인플루엔자 확산 예측을 위한 시공간 모델링 및 영향 요인 분석). *Annual Conference of the Korea Environmental Economics Association*, Aug. 2024.

[6] "Automation of Algal Bloom Prediction Through a Deep Learning-Based MLOps Pipeline: AI Model Serving for Policy Utilization" (딥러닝 기반 MLOps 파이프라인을 통한 녹조 예측 자동화: 정책적 활용을 위한 인공지능 모델 서빙 접근). *Summer Conference of the Korea Society for Innovation Management & Economics (KOSIME)*, Jul. 2024.

[7] "(Session for Emerging Researchers) Development and Application of AI for Environmental and Infectious Disease Predictions" ((신진연구자 세션) 환경 및 감염병 예측을 위한 인공지능의 개발과 실제). *Korea Software Congress*, Dec. 2023.

[8] "Explainable Deep Learning Integration for Algal Bloom Prediction and MLOps Implementation" (설명 가능한 녹조 통합 딥러닝 예측과 MLOps). *Spring Conference of the Korean Society of Analytical Science*, May 2023.

[9] "Fine Dust Prediction Using Convolutional Neural Network Deep Learning" (컨벌루션 신경망 딥러닝을 활용한 미세먼지 예측). *Summer Conference of the Korea Environmental Economics Association*, Aug. 2018.

[10] "Stock Price Prediction and Profitability Evaluation Using Deep Learning Analysis of Financial Big Data" (재무 빅데이터의 딥러닝 분석을 통한 주가예측 및 수익률 평가). *Summer Conference of the Korea Society for Innovation Management & Economics (KOSIME)*, Jul. 2018.

[11] "Risk Prediction for Fine Dust Using Deep Learning Technology" (딥러닝 기술을 활용한 미세먼지 리스크 예측). *Summer Conference of the Korea Society for Innovation Management & Economics (KOSIME)*, Jun. 2017.

[12] "Complex System Analysis of Earthquake Big Data: Focusing on Regional Concentration Phenomena" (지진 빅데이터를 활용한 복잡계 현상 분석:지진의 지역 집중화 현상을 중심으로). *Summer Conference of the Korea Regional Information Society*, Jun. 2017.

[13] "Application of Deep Learning Technology in Environmental Research: Predicting Eco-Friendly Consumption Indices" (딥러닝 기술의 환경 연구에의 적용: 친환경 소비 지표 예측을 중심으로). *Summer Conference of the Korea Society for Innovation Management & Economics (KOSIME)*, Jun. 2016.

[14] "A Farewell to Arms? Patent War in the Smartphone Industry." *Joint Economics Conference*, Feb. 2015.

PATENTS

[1] **"Method for Predicting Air Quality Using Artificial Intelligence."** Korean Patent No. 10-2743462, registered Dec. 2024 (Application No. 10-2022-0015727, filed Feb. 2022).

[2] **"Improved Device and Method for Predicting Fine Dust Concentration."** Korean Patent No. 10-2984494, registered Jun. 2026 (Application No. 10-2022-0082928, filed Jul. 2022).

[3] **"Method and Device for Predicting Infectious Diseases Using Artificial Intelligence."** Korean Patent No. 10-2966845, registered May 2026 (Application No. 10-2022-0132078, filed Oct. 2022).

[4] **"Method and Device for Predicting Algal Bloom Occurrences Using an Artificial Intelligence Model."** Korean Patent No. 10-2922243, registered Jan. 2026 (Application No. 10-2022-0172370, filed Dec. 2022).

[5] **"Method and Device for Modeling Generation for Time Series Prediction."** Patent Application No. 10-2023-0010001, filed Jan. 2023.

HONORS AND AWARDS

- **Outstanding Abstract Award (우수초록상)** (Sep. 11, 2026)
  2026 Fall Conference of the Korean Society for Zoonoses
  H. Jeon and **D. Lee** (corresponding author). Certificate issued in the name of the Society President.
- **President's Award of the Korea Environment Institute (한국환경연구원장상)** (2024)
  Environmental Data Analysis Contest: Development of Multimodal LLM Interactive AI
- **Outstanding Researcher Award (Excellence in Academic Paper Category)** (2019)
  With the President's Commendation at Tech University of Korea
- **Presidential Award for the Korea Young Talent Award (대한민국 인재상 대통령상)** (2008)
  Presented by the President of South Korea

RESEARCH METHODS AND TECHNICAL SKILLS

- **Research methods:** Computational modeling, uncertainty quantification, explanation analysis, transfer learning, and edge-case validation; collaborative research using survey and choice data.
- **Programming and analysis:** Python, R, Java, SQL; PyTorch, TensorFlow, scikit-learn, NumPy, Pandas; SPSS and Stata.
- **Research software:** Docker, GitHub Actions, REST APIs, Matplotlib, and Streamlit.